

MAT 350: Linear Transformations and their Matrix Representations

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Warm-Up Problems

Complete the following warm-up problems to re-familiarize yourself with concepts we'll be leveraging today.

1. Consider the matrix $A = \begin{bmatrix} -2 & 0 & 5 \\ 1 & 1 & 3 \end{bmatrix}$ and the vector $\vec{x} = \begin{bmatrix} 5 \\ -4 \\ 1 \end{bmatrix}$. Compute the product $A\vec{x}$.
2. Consider the matrix $A = \begin{bmatrix} -1 & 0 & 1 \\ 5 & 1 & -1 \\ 0 & 0 & 3 \end{bmatrix}$ and the vector $\vec{b} = \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix}$. Determine whether solutions to the matrix equation $A\vec{x} = \vec{b}$ exists. Describe the geometry of the solution space.
3. Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by $f(\vec{x}) = \begin{bmatrix} -2 & 0 & 5 \\ 1 & 1 & 3 \end{bmatrix} \vec{x}$. Evaluate $f\left(\begin{bmatrix} 5 \\ -4 \\ 1 \end{bmatrix}\right)$
4. Given the same function f defined in part (3), compute both $f\left(\begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}\right)$ and $f\left(-3 \cdot \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}\right)$. What, if anything, do you notice?
5. Using the same function f defined in part (3) again, compute $f\left(\begin{bmatrix} 5 \\ -4 \\ 1 \end{bmatrix}\right) + (-3)\begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}$. What, if anything, do you notice now?

Reminders and Today's Goal

- Matrix-vector multiplication ($A\vec{x}$) is defined for matrices and vectors of compatible sizes.
 - The number of columns in the matrix A must match the number of entries in the vector $\vec{x}$.
 - The result of multiplying an $m \times n$ matrix by an $n \times 1$ vector is an $m \times 1$ vector
 - Multiplying an n -element vector by an $n \times n$ matrix moves that vector within the space $\mathbb{R}^n$
 - Multiplying an n -element vector by an $m \times n$ matrix (with $m \neq n$) transforms the vector from $\mathbb{R}^n$ to $\mathbb{R}^m$
- In the warm-up problems (specifically 3 - 5), you considered a function defined by matrix multiplication.
- In fact, this isn't the first time we've considered such functions.
- Each time we've encountered *matrix equations*, we've brought up this notion.

Reminders and Today's Goal

Goals for Today: After today's discussion, you should be able to...

- Define *linear transformation* as a special class of function from $\mathbb{R}^n$ to $\mathbb{R}^m$.
- Identify characteristics that linear transformations must possess.
- Determine whether a transformation is *linear*.
- Write the matrix corresponding to a linear transformation.

Properties of Matrix Multiplication

In order to develop and define the notion of a matrix transformation, we'll need to formalize some properties of matrix multiplication that we haven't explicitly mentioned yet.

For the following, consider that A and B are matrices, $\vec{x}$ and $\vec{y}$ are vectors, and c and d are scalars.

- $A(c\vec{x}) = cA\vec{x}$
- $A(\vec{x} + \vec{y}) = A\vec{x} + A\vec{y}$
- $(A + B)\vec{x} = A\vec{x} + B\vec{x}$

Combining the first two bullet points above to notice that $A(\vec{x} + c\vec{y}) = A\vec{x} + cA\vec{y}$ will be particularly useful.

Additional Reminders: Since we're talking about matrix multiplication, it's worth reminding ourselves that matrix multiplication is not commutative. That is, it is not generally the case that AB and BA are the same, and $A\vec{x}$ is not the same as $\vec{x}A$.

Matrix Transformations

- Given an $m \times n$ matrix A , we can define a transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ by $T(\vec{x}) = A\vec{x}$.
- Such a transformation will send vectors from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Example: Consider the transformation $T(\vec{x}) = A\vec{x}$, where the matrix

$$A = \begin{bmatrix} 2 & -3 \\ 1 & 0 \\ -4 & 5 \end{bmatrix}.$$

- Since the matrix A is a 3×2 matrix, the transformation T takes vectors from $\mathbb{R}^2$ (its domain) to $\mathbb{R}^3$ (its codomain).

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- We can come up with a vector notation for the transformation as well.

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2 & -3 \\ 1 & 0 \\ -4 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

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That is, given an input vector $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$,

we have $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 3x_1 - 3x_2 \\ x_1 \\ -4x_1 + 5x_2 \end{bmatrix}$,

Linear Transformations

We'll return to *matrix transformations* shortly, but let's define a general *linear transformation* first.

Definition (Linear Transformation): A transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is called a *linear transformation* if, for any vectors $\vec{x}, \vec{y} \in \mathbb{R}^n$ and any scalar c , the following property holds

$$T(\vec{x} + c\vec{y}) = T(\vec{x}) + cT(\vec{y})$$

Note that any *matrix transformation* is a *linear transformation* since, if $T(\vec{x}) = A\vec{x}$, then

$$T(\vec{x} + c\vec{y})$$

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$$\begin{aligned} T(\vec{x} + c\vec{y}) &= A(\vec{x} + c\vec{y}) \\ &= A\vec{x} + A(c\vec{y}) \end{aligned}$$

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In fact, it can be shown that, for any general linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, there exists an $m \times n$ matrix A such that $T(\vec{x}) = A\vec{x}$

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Note that any *matrix transformation* is a *linear transformation* since, if $T(\vec{x}) = A\vec{x}$, then

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In fact, it can be shown that, for any general linear transformation

$T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, there exists an $m \times n$ matrix A such that $T(\vec{x}) = A\vec{x}$

That is, every *linear transformation* is also a *matrix transformation*.

Writing Linear Transformations as Matrix Transformations

We won't prove that every *linear transformation* is also a *matrix transformation* in complete generality.

We'll do it here through an example, showing a strategy to discover the matrix associated with any *linear transformation*.

Example: Consider the linear transformation T

$$T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{bmatrix} x_1 - 3x_3 + x_4 \\ x_3 - 3x_4 \end{bmatrix}.$$

Writing Linear Transformations as Matrix Transformations

Example: Consider the linear transformation $T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \right) = \begin{bmatrix} x_1 - 3x_3 + x_4 \\ x_3 - 3x_4 \end{bmatrix}$.

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$$\begin{aligned} T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}\right) &= \begin{bmatrix} x_1 - 3x_3 + x_4 \\ x_3 - 3x_4 \end{bmatrix} \\ &= \begin{bmatrix} x_1 \\ 0 \end{bmatrix} + \begin{bmatrix} -3x_3 \\ x_3 \end{bmatrix} + x_4 \begin{bmatrix} x_4 \\ -3x_4 \end{bmatrix} \end{aligned}$$

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$$\begin{aligned} T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \right) &= \begin{bmatrix} x_1 - 3x_3 + x_4 \\ x_3 - 3x_4 \end{bmatrix} \\ &= \begin{bmatrix} x_1 \\ 0 \end{bmatrix} + \begin{bmatrix} -3x_3 \\ x_3 \end{bmatrix} + x_4 \begin{bmatrix} x_4 \\ -3x_4 \end{bmatrix} \\ &= x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 1 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ -3 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & -3 & 1 \\ 0 & 0 & 1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \end{aligned}$$

That is, $T(\vec{x}) = A\vec{x}$
where

$$A = \begin{bmatrix} 1 & 0 & -3 & 1 \\ 0 & 0 & 1 & -3 \end{bmatrix}.$$

Examples to Try # 1

Before we move forward, here are a couple of example problems for you to verify your understanding with. For each example, consider the transformation $T(\vec{x}) = A\vec{x}$

where $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.

1. Evaluate $T\left(\begin{bmatrix} 1 \\ -2 \end{bmatrix}\right)$

2. Evaluate $T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right)$

3. Evaluate $T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)$

4. Is there a vector $\vec{x}$ such that $T(\vec{x}) = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$

5. Write $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right)$ as a linear combination of vectors with weights x_1 and x_2

Properties of Linear Transformations

Now that we've identified that any *linear transformation* corresponds to some *matrix transformation*, we can investigate properties of *linear transformations*.

All of the upcoming properties result from properties of matrix multiplication.

Property 1 (Transformation of $\vec{0}$): $T(\vec{0}) = \vec{0}$

$$\begin{aligned}\text{Proof. } T(\vec{0}) &= A\vec{0} \\ &= \vec{0}\end{aligned}$$

Properties of Linear Transformations

Property 2 (Linearity): Consider the vectors $\vec{x_1}$ and $\vec{x_2}$, along with the scalar c , then
 $T(\vec{x_1} + c\vec{x_2}) = T(\vec{x_1}) + cT(\vec{x_2})$

$$\begin{aligned}\text{Proof. } T(\vec{x_1} + c\vec{x_2}) &= A(\vec{x_1} + c\vec{x_2}) \\ &= A\vec{x_1} + A(c\vec{x_2}) \\ &= A\vec{x_1} + cA\vec{x_2} \\ &= T(\vec{x_1}) + cT(\vec{x_2})\end{aligned}$$

Properties of Linear Transformations

New Ideas/Notations Needed: We need a few new ideas and notations before listing out additional properties of linear transformations.

Consider the following special vectors, which we'll investigate in greater detail soon.

$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}, \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}, \dots, \vec{e}_n = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

- The vectors $\vec{e}_i$ point in the directions of the usual axes in $\mathbb{R}^n$.
 - For clarity, consider $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ in $\mathbb{R}^2$.

Properties of Linear Transformations

Property 3 (Structure of the Matrix A): If $A = \begin{bmatrix} \vec{v}_1 & \vec{v}_2 & \cdots & \vec{v}_n \end{bmatrix}$, then $T(\vec{e}_i) = \vec{v}_i$.
This means that, if $T(\vec{x}) = A\vec{x}$, then

$$A = \begin{bmatrix} T(\vec{e}_1) & T(\vec{e}_2) & \cdots & T(\vec{e}_n) \end{bmatrix}$$

Proof. Omitted for this course...

This property is extremely useful for two reasons. If you know the result of $T(\vec{e}_i)$ for all i ,

1. you can construct the matrix corresponding to the linear transformation.
2. you can evaluate $T(\vec{x})$ for any $\vec{x}$ in the domain of T .

Preview: This property will generalize and will be even more impactful later in our course.

Completed Example # 2

Example: Consider the transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ satisfying the following:

$$T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ -3 \\ 1 \\ 1 \end{bmatrix} \quad T\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \\ -2 \\ 5 \end{bmatrix} \quad T\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \\ 4 \\ -1 \end{bmatrix} \quad \text{Evaluate } T\left(\begin{bmatrix} -4 \\ 2 \\ -3 \end{bmatrix}\right).$$

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$$T\left(\begin{bmatrix} -4 \\ 2 \\ -3 \end{bmatrix}\right) = T\left(-4\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 2\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - 3\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right)$$

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$$\begin{aligned} T\left(\begin{bmatrix} -4 \\ 2 \\ -3 \end{bmatrix}\right) &= T\left(-4\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 2\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - 3\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) \\ &= T\left(-4\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) + T\left(2\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) + T\left(-3\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) \\ &= -4T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) + 2T\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) + (-3)T\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) \end{aligned}$$

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Completed Example # 2

Example: Consider the transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ satisfying the following:

$$T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ -3 \\ 1 \\ 1 \end{bmatrix} \quad T\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \\ -2 \\ 5 \end{bmatrix} \quad T\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \\ 4 \\ -1 \end{bmatrix} \quad \text{Evaluate } T\left(\begin{bmatrix} -4 \\ 2 \\ -3 \end{bmatrix}\right).$$

$$\begin{aligned} T\left(\begin{bmatrix} -4 \\ 2 \\ -3 \end{bmatrix}\right) &= -4T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) + 2T\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) + (-3)T\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) \\ &= -4\begin{bmatrix} 2 \\ -3 \\ 1 \\ 1 \end{bmatrix} + 2\begin{bmatrix} 1 \\ 0 \\ -2 \\ 5 \end{bmatrix} + (-3)\begin{bmatrix} 0 \\ 1 \\ 4 \\ -1 \end{bmatrix} \end{aligned}$$

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$$\begin{aligned} T\left(\begin{bmatrix} -4 \\ 2 \\ -3 \end{bmatrix}\right) &= -4T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) + 2T\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) + (-3)T\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) \\ &= -4\begin{bmatrix} 2 \\ -3 \\ 1 \\ 1 \end{bmatrix} + 2\begin{bmatrix} 1 \\ 0 \\ -2 \\ 5 \end{bmatrix} + (-3)\begin{bmatrix} 0 \\ 1 \\ 4 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} -8 \\ 12 \\ -4 \\ -4 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \\ -4 \\ 10 \end{bmatrix} + \begin{bmatrix} 0 \\ -3 \\ -12 \\ 3 \end{bmatrix} \end{aligned}$$

Completed Example # 2

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$$T\left(\begin{bmatrix} -4 \\ 2 \\ -3 \end{bmatrix}\right) = \begin{bmatrix} -8 \\ 12 \\ -4 \\ -4 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \\ -4 \\ 10 \end{bmatrix} + \begin{bmatrix} 0 \\ -3 \\ -12 \\ 3 \end{bmatrix}$$

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$$T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ -3 \\ 1 \\ 1 \end{bmatrix} \quad T\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \\ -2 \\ 5 \end{bmatrix} \quad T\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \\ 4 \\ -1 \end{bmatrix} \quad \text{Evaluate } T\left(\begin{bmatrix} -4 \\ 2 \\ -3 \end{bmatrix}\right).$$

$$\begin{aligned} T\left(\begin{bmatrix} -4 \\ 2 \\ -3 \end{bmatrix}\right) &= \begin{bmatrix} -8 \\ 12 \\ -4 \\ -4 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \\ -4 \\ 10 \end{bmatrix} + \begin{bmatrix} 0 \\ -3 \\ -12 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} -6 \\ 9 \\ -20 \\ 9 \end{bmatrix} \end{aligned}$$

Completed Example # 2

Example: Consider the transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ satisfying the following:

$$T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ -3 \\ 1 \\ 1 \end{bmatrix} \quad T\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \\ -2 \\ 5 \end{bmatrix} \quad T\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \\ 4 \\ -1 \end{bmatrix} \quad \text{Evaluate } T\left(\begin{bmatrix} -4 \\ 2 \\ -3 \end{bmatrix}\right).$$

$$\begin{aligned} T\left(\begin{bmatrix} -4 \\ 2 \\ -3 \end{bmatrix}\right) &= \begin{bmatrix} -8 \\ 12 \\ -4 \\ -4 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \\ -4 \\ 10 \end{bmatrix} + \begin{bmatrix} 0 \\ -3 \\ -12 \\ 3 \end{bmatrix} \\ &= \boxed{\begin{bmatrix} -6 \\ 9 \\ -20 \\ 9 \end{bmatrix}} \end{aligned}$$

Superposition Principle

In that recent example, we made use of an extension of the linearity property of linear transformations, which we'll state explicitly below.

Linear Transformations and Linear Combinations: For any linear transformation T , we have that

$$T \left(c_1 \vec{v}_1 + c_2 \vec{v}_2 + \cdots + c_n \vec{v}_n \right) = c_1 T \left(\vec{v}_1 \right) + c_2 T \left(\vec{v}_2 \right) + \cdots + c_n T \left(\vec{v}_n \right)$$

The extension above gives rise to the **superposition principle** from engineering and physics.

Superposition Principle: If we consider $\vec{v}_1, \vec{v}_2, \cdots, \vec{v}_n$ as signals entering a system and $T \left(\vec{v}_1 \right), T \left(\vec{v}_2 \right), \cdots, T \left(\vec{v}_n \right)$ as the responses of the system to each signal, then the response to a linear combination of the signals will be a linear combination of the responses to the individual signals with the same weights.

Summary

Homework

Start Homework 6
on MyOpenMath

Next Time...

Geometry of Matrix Transformations